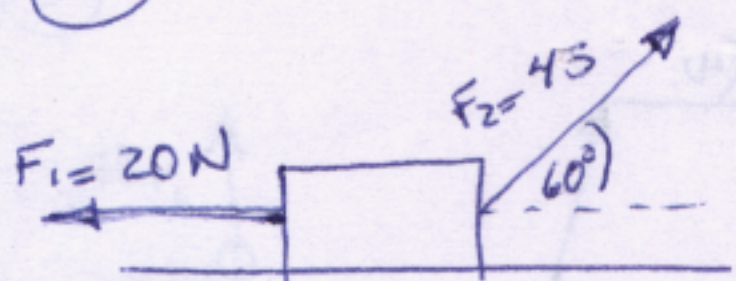


①



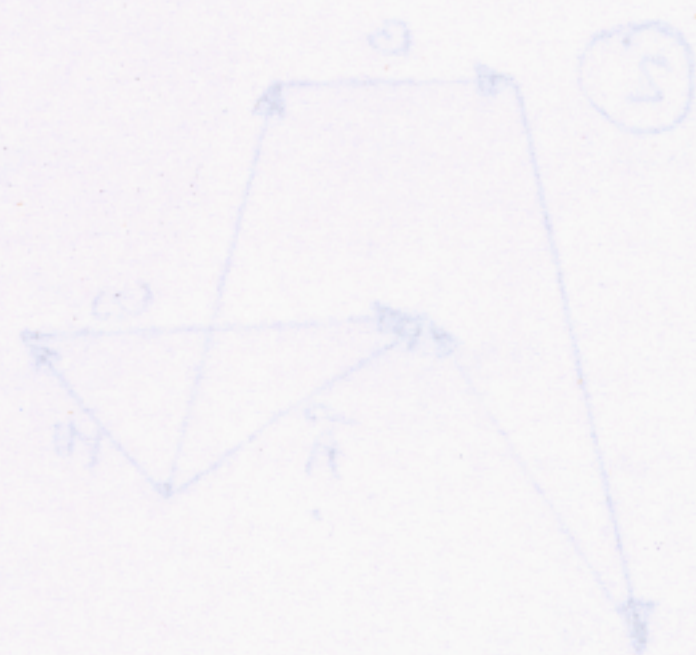
	X	Y
F_1	-20	0
F_2	$45 \cos 60$	$45 \sin 60$
R	2,5	$22,5 \sqrt{3}$

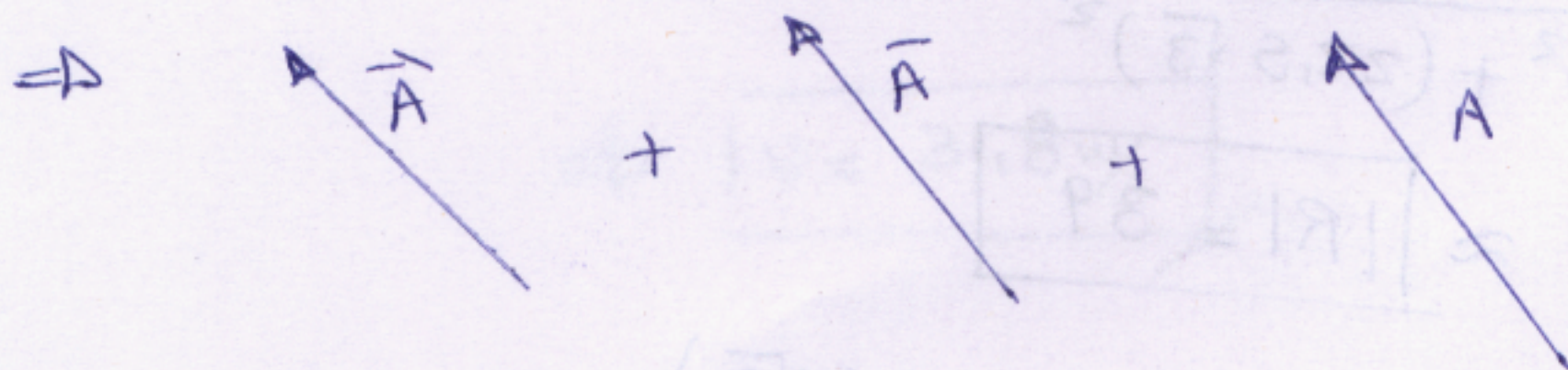
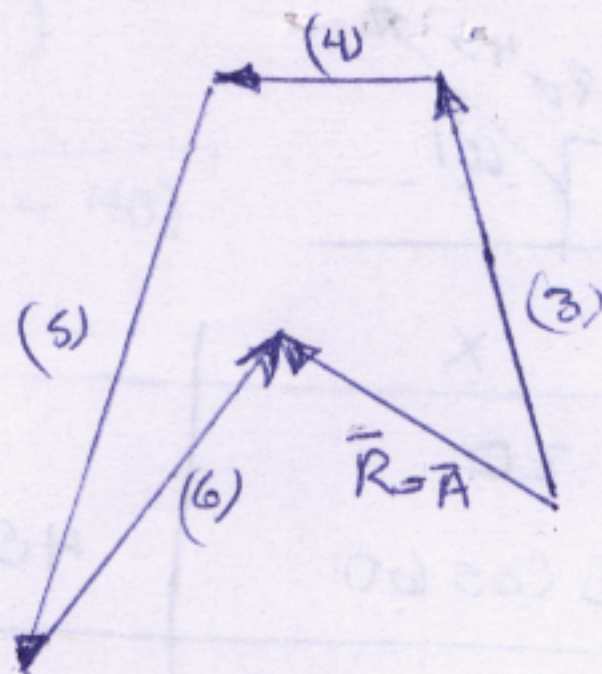
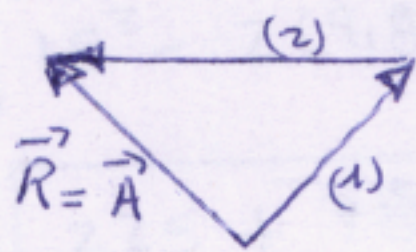
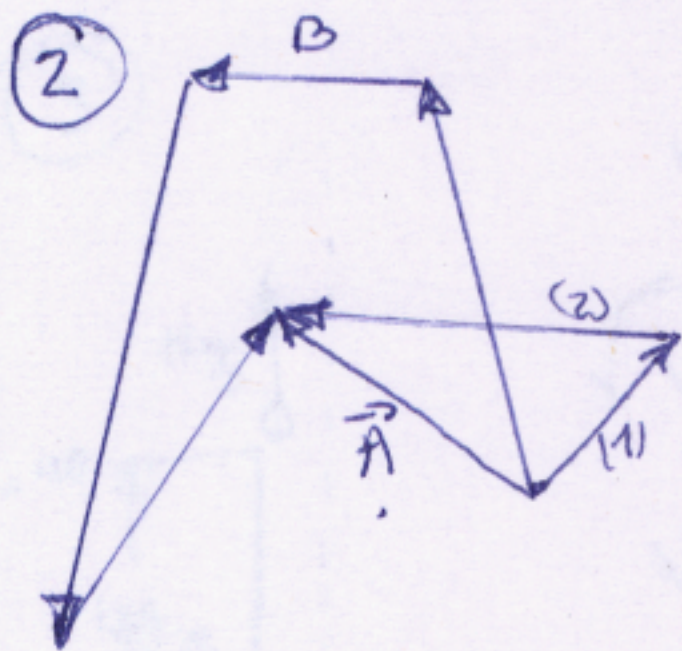
$$|R| = \sqrt{(2,5)^2 + (22,5\sqrt{3})^2}$$

$$|R| = 39,05 \approx \boxed{|R| = 39}$$

$$\tan \theta = \frac{R_y}{R_x} \Rightarrow \theta = \tan^{-1} \left(\frac{22,5\sqrt{3}}{2,5} \right)$$

$$\theta = 86,33 \approx \boxed{\theta = 86^\circ}$$

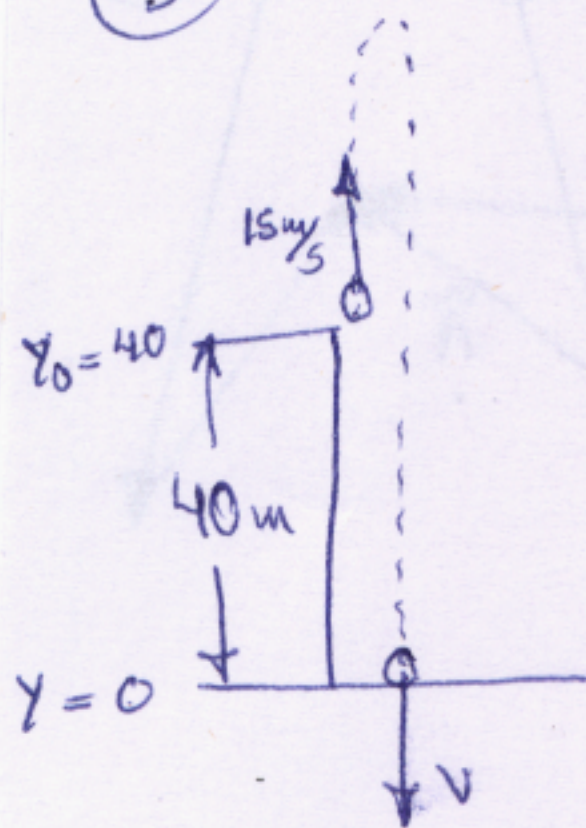




$$R = 3A = R = 3(10)$$

$$\boxed{|R| = 30}$$

3



$$V^2 = V_0^2 - 2g(y - y_0)$$

$$V = \sqrt{V_0^2 - 2g(y - y_0)}$$

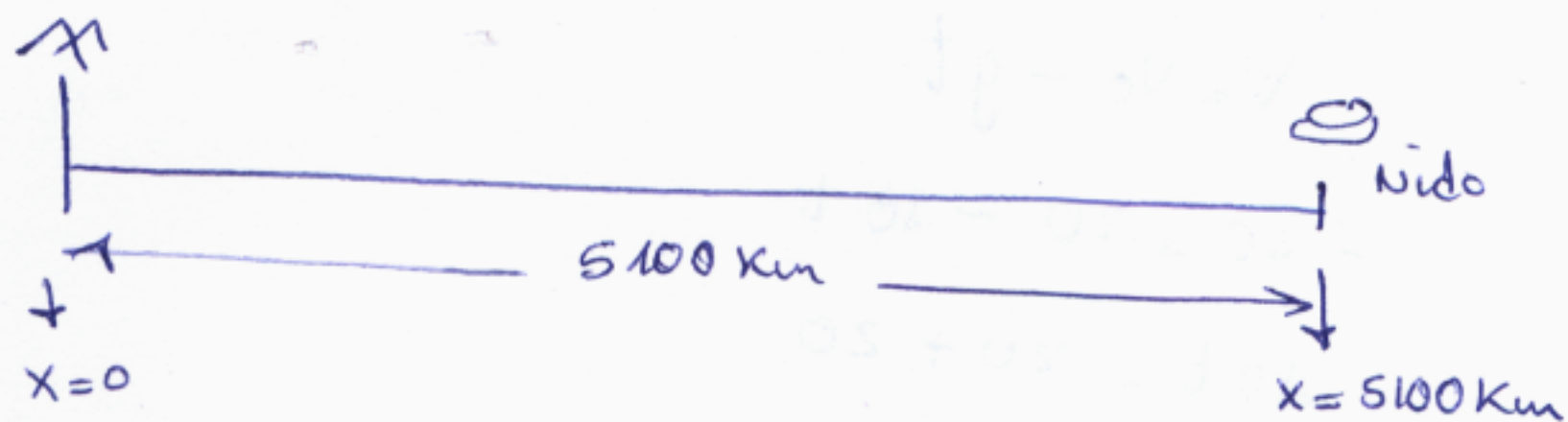
$$V = \sqrt{225 + 784}$$

$$V = \sqrt{1009}$$

$$V = 31,76 \approx 31,8 \text{ m/s}$$

$$\Rightarrow \boxed{V = 31,8 \text{ m/s}}$$

4



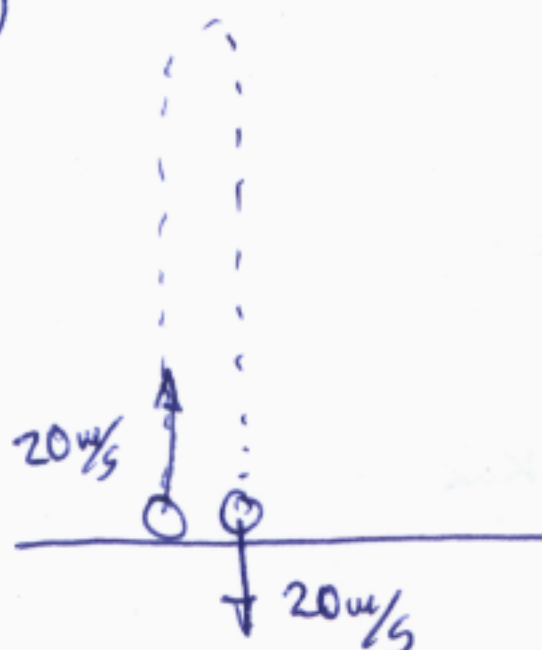
$$X = 5100 \text{ Km} \times \frac{1000 \text{ m}}{1 \text{ Km}} \Rightarrow X = 5100000 \text{ m}$$

$$t = 12,4 \text{ dias} \times \frac{24 \text{ h}}{1 \text{ dia}} \times \frac{3600 \text{ s}}{1 \text{ h}} \Rightarrow t = 1071360 \text{ s}$$

$$V = \frac{\Delta X}{\Delta t} = \frac{X - X_0}{t} = \frac{5100000 - 0}{1071360} = \frac{5100000}{1071360}$$

$$\boxed{V_m = 4,76 \text{ m/s}}$$

5



$$V = V_0 - gt$$
$$-20 = 20 - 10t$$

$$10t = 20 + 20$$

$$t = \frac{40}{10}$$

$$t = 4\text{ s}$$